TRY HACK ME

MEMORY ANALYSIS INTRODUCTION

NAME: ALLAN NJUGUNA NGUGI

COURSE: COMPUTER FORENSICS AND INVESTIGATION

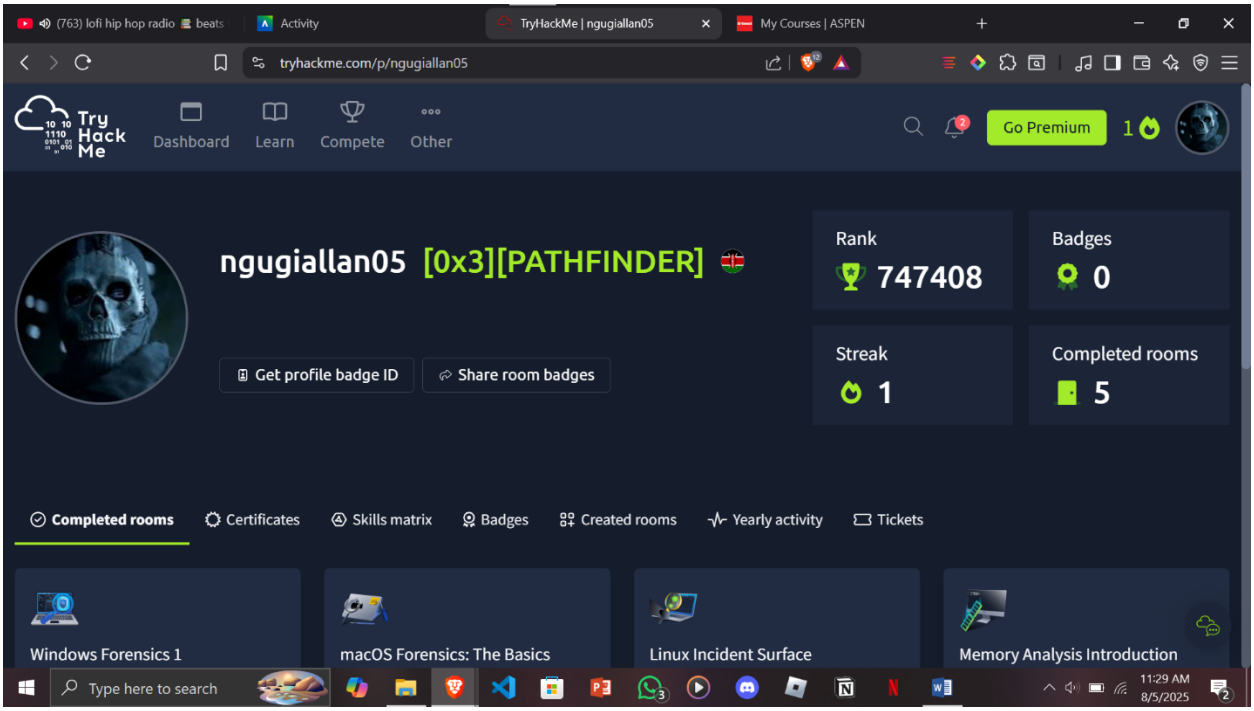
ASSIGNMENT: LAB ASSIGNMENT

SUPERVISOR: PROF. RICHARD KARANJA

TABLE OF CONTENTS

PROFILE 2
Task 1: Introduction 3
Key Takeaway 3
Task 2: Volatile Memory 5
Key Takeaway 5
Task 3: Memory Dumps 7
Key Takeaway 8
Task 4: Memory Analysis Attack Fingerprints 9
Key Takeaway 9
Task 5: Practical 10
Key Takeaway 11
Task 6: Conclusion 12
References 13

PROFILE

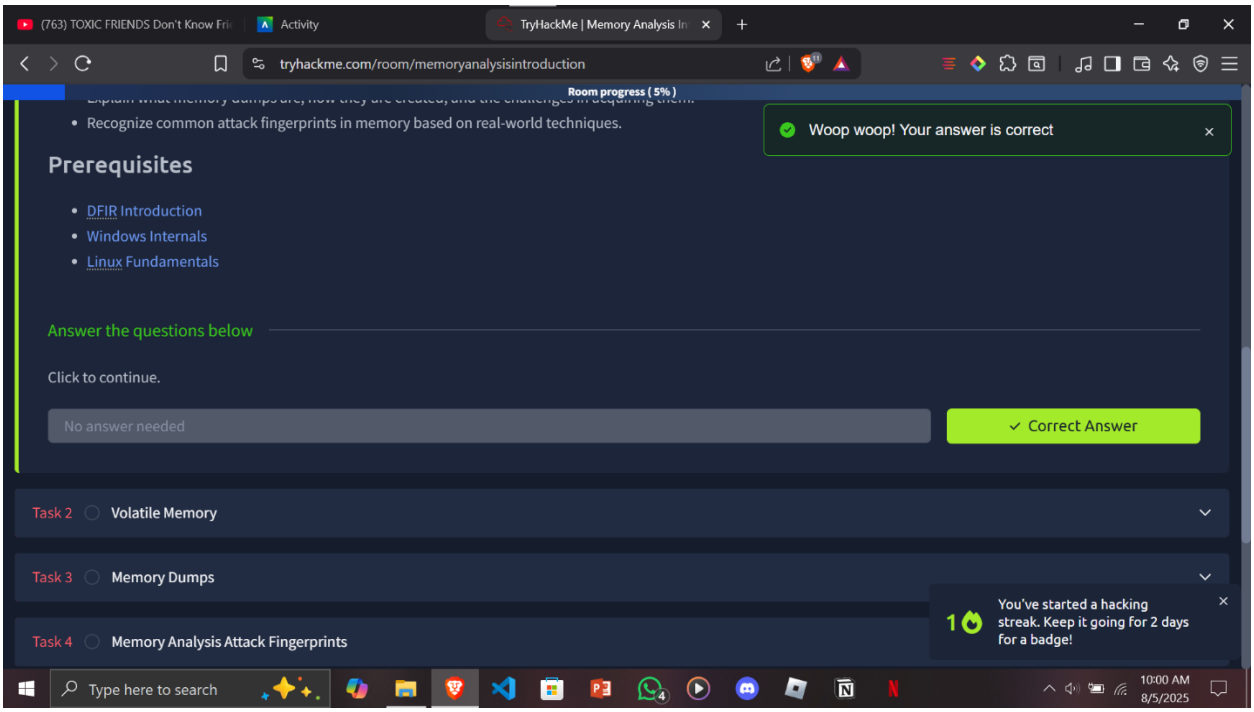

(763) lofi hip hop radio beats
Activity
TryHackMe | ngugiallan05
My Courses | ASPEN
tryhackme.com/p/ngugiallan05
Try Hack Me
Dashboard
Learn
Compete
Other
Go Premium
1
ngugiallan05 [0x3][PATHFINDER]
Get profile badge ID
Share room badges
Rank
747408
Badges
0
Streak
1
Completed rooms
5
Completed rooms
Certificates
Skills matrix
Badges
Created rooms
Yearly activity
Tickets
Windows Forensics 1
macOS Forensics: The Basics
Linux Incident Surface
Memory Analysis Introduction
Type here to search
11:29 AM
8/5/2025

Task 1: Introduction

We will examine how memory analysis aids cyber security investigations in this room, particularly from the viewpoint of a blue team. We'll discover how information stored in RAM can disclose user activities, ongoing threats, and artifacts that are erased upon shutdown.

The fundamentals of volatile memory and its relationship to disk forensics will be covered first. Next, we'll examine the creation and usage of memory dumps, potential indications of memory attacks, and the situations in which this kind of analysis is most helpful.

We will be able to see forensic evidence and memory structure later in the room thanks to an interactive part. By the end, we'll know when and how to use memory analysis in practical research.



Key Takeaway

Memory analysis is an important aspect of digital forensics and incident response that is most beneficial in the investigation of live systems while or soon after a cyber-attack has occurred. As opposed to typical disk-based forensics, memory analysis is focused on volatile information that exists only when the computer is online. This project provides context by showing how

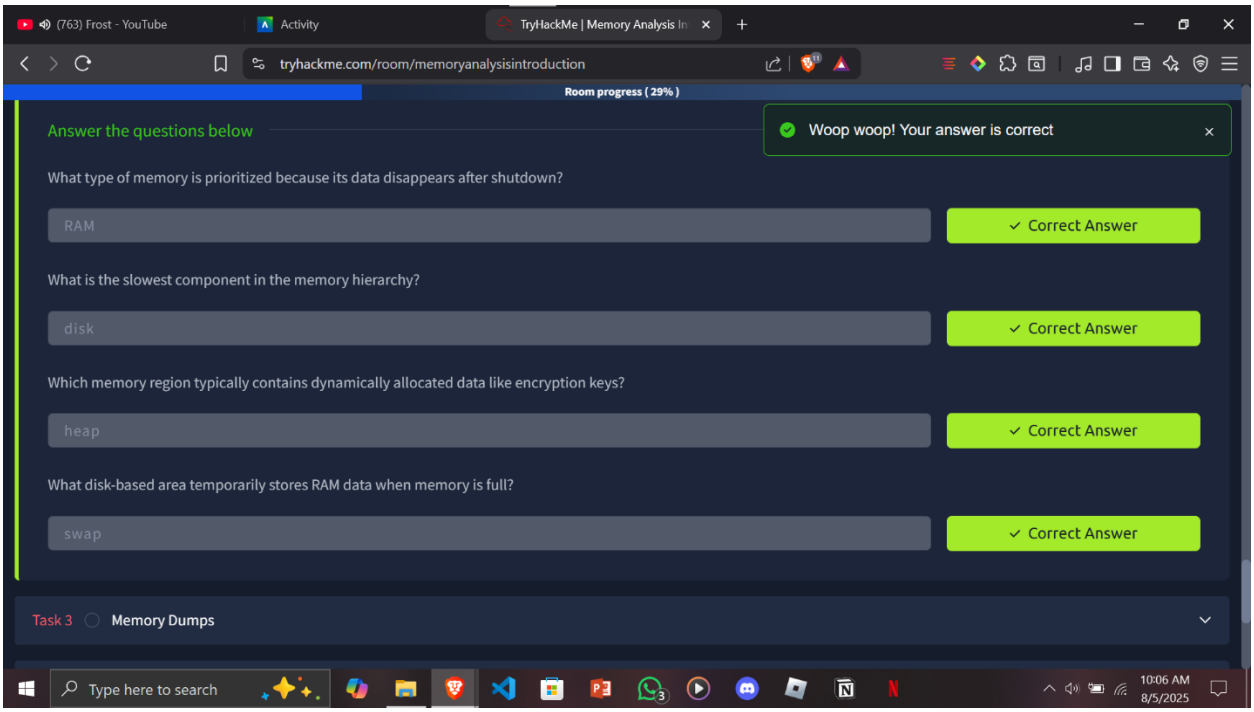
important RAM data is in uncovering operating processes, network traffic, operating malware, and user interactions in real time.

Task 2: Volatile Memory

This task offers information based off on volatile memory and its importance in investigations and, more generally, digital forensics. While a computer is operating, system and user-level data are kept in what is known as volatile memory. This data is lost when the system is turned off or restarted. The Random Access Memory (RAM) is a typical location for this kind of memory.

Everything is momentarily stored in RAM, including encrypted data, open files, and active activities. Investigators frequently give priority to acquiring RAM as early as possible during an inquiry because this data is only available when the machine is operating.

After carefully going through the task, the following questions were answered correctly:



Key Takeaway

Volatile memory, or RAM, has sensitive system and user-level data that is lost during system shutdown, so it is crucial to maintain it early in an investigation. Current running processes, open files, encryption keys, and even potential malware payloads are stored there. Knowledge of memory architecture—regions such as the stack, heap,

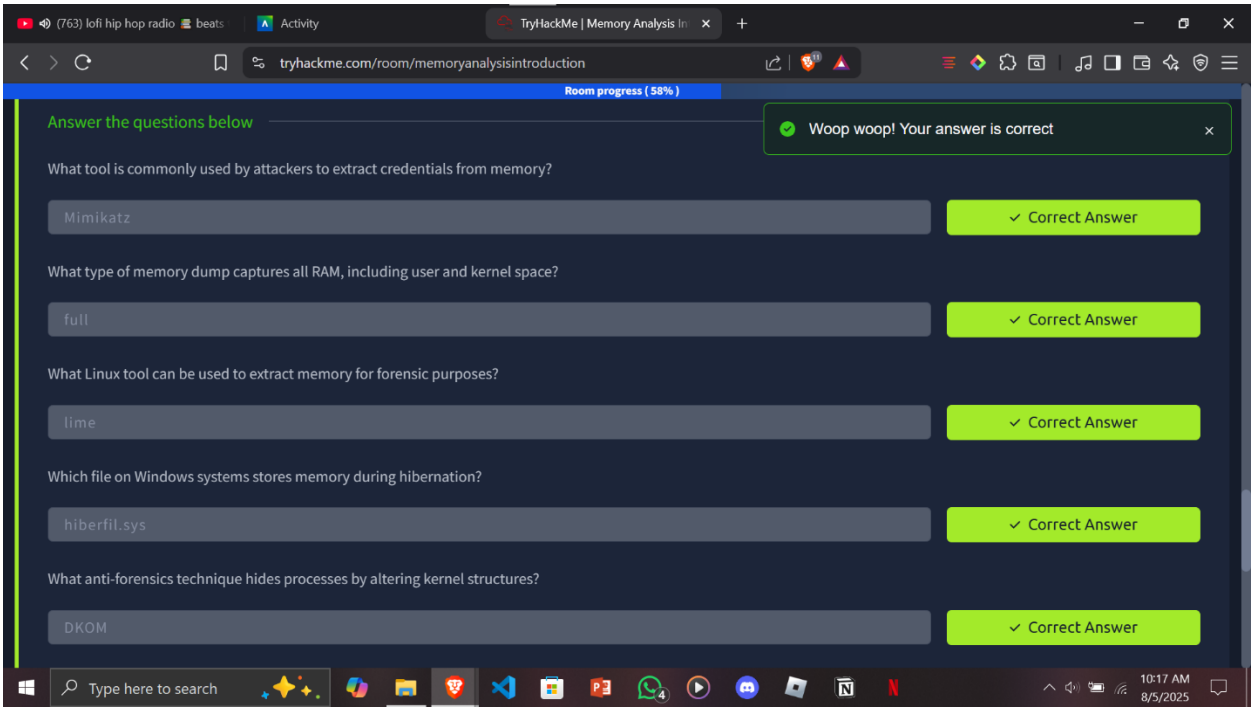
and various segments such as .text and .data—allows investigators to better locate forensic evidence.

Task 3: Memory Dumps

We examine the definition of a memory dump and its application to forensic procedures in this task. An image of a system's RAM taken at a particular moment in time is called a memory dump. It records everything that is kept in volatile memory, including active sessions, processes that are currently executing, network activities, and occasionally even private information like login passwords.

Memory dumps are frequently utilized in threat hunting, malware investigations, and forensic analysis. These snapshots are examined by security teams to determine what was operating on a system and to find any unusual or suspicious activities. Memory dumps are a crucial defensive focus since attackers and red teamers frequently utilize tools like Mimikatz to harvest credentials straight from memory.

Following the completion of the task, there were question answered correctly. This is shown in the image below:



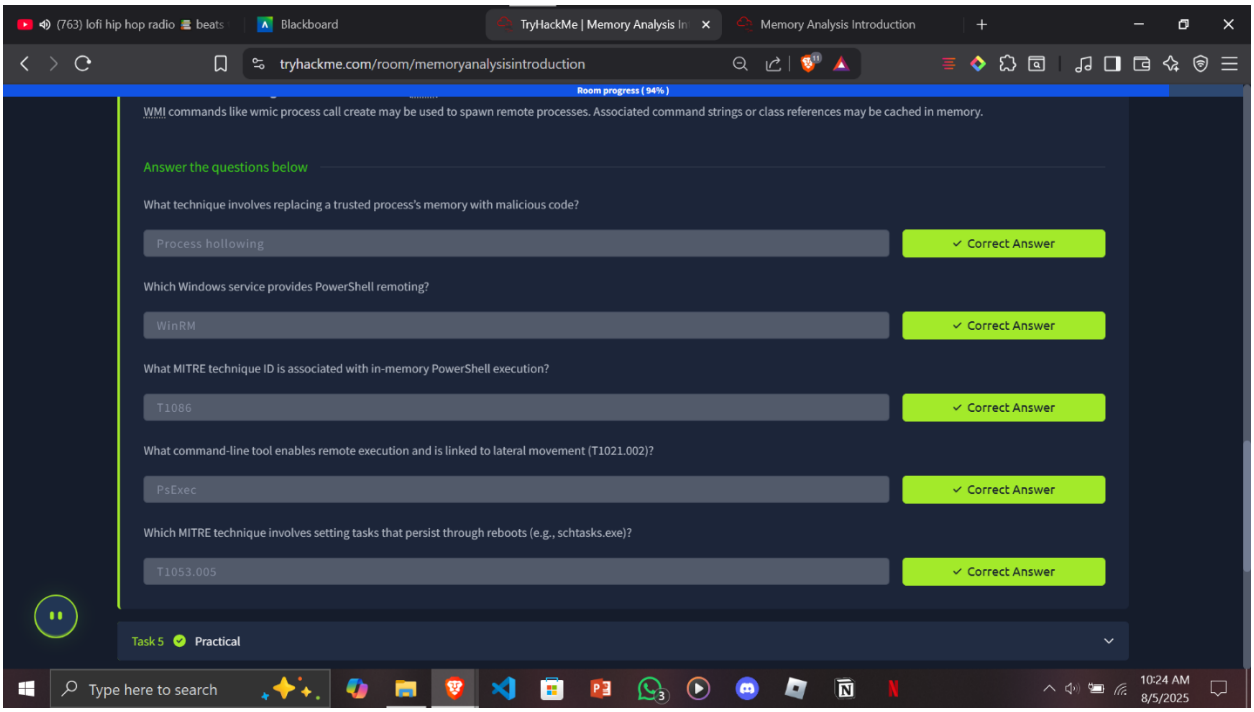
Key Takeaway

Memory dumps are snapshots of the RAM of a system at a specific moment and are important in preserving volatile data for analysis. Dumps are obtained using tools like WinPmem, FTK Imager, or crash dumps and may include full memory, single processes, or even swap files that hold ephemeral memory information. Analysts must also be cognizant of anti-forensic measures like process hiding via Direct Kernel Object Manipulation (DKOM) and the utilization of encrypted payloads that render analysis challenging.

Task 4: Memory Analysis Attack Fingerprints

Memory analysis is particularly effective in identifying dangers that might not be detected by disk traces. Logs and file systems provide long-term proof, but memory shows what is happening at the moment, which makes it crucial for detecting recent or ongoing attacks.

Upon completion of the task, the questions provided were answered correctly. This is shown in the image below:

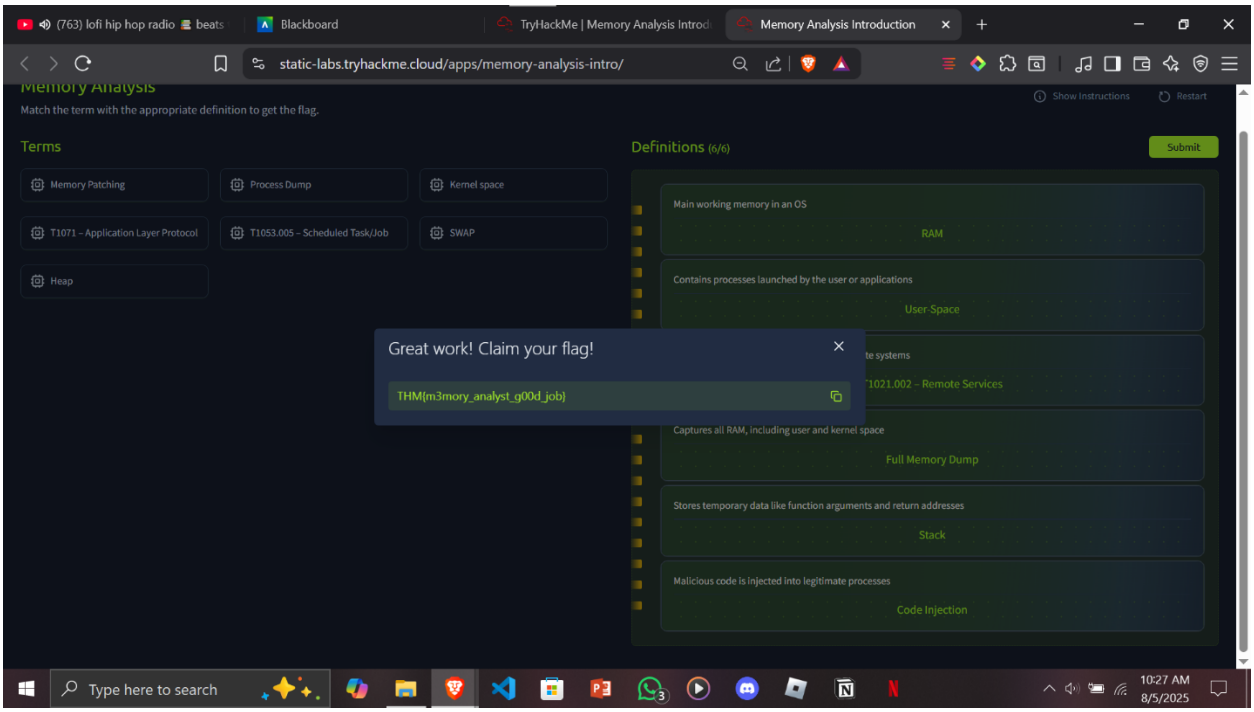


Key Takeaway

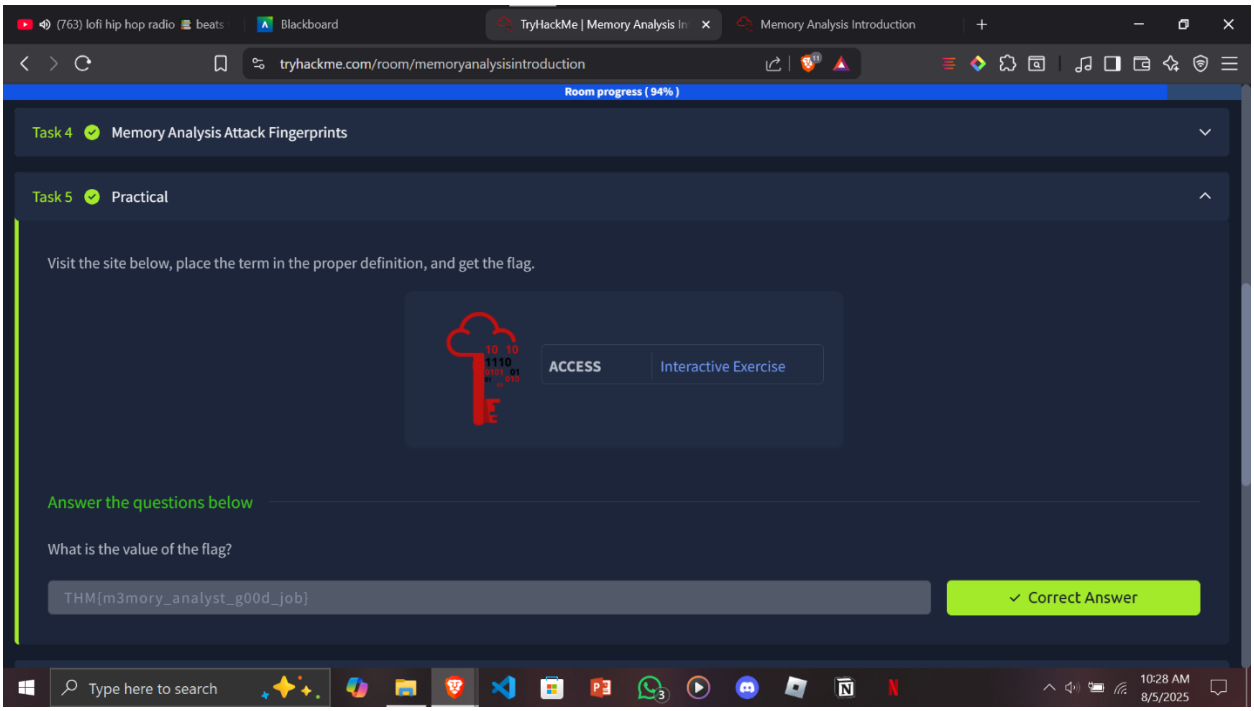
RAM analysis enables the identification of advanced threat techniques like fileless malware, process injection, and rootkits. Analysts are able to detect anomalies like unsigned processes, execution from non-standard memory locations, or unusual scheduled tasks reflecting persistence mechanisms. Correlating these behaviors with MITRE ATT&CK techniques—PowerShell-based attacks (T1086), malicious services (T1543.003), or remote access via WinRM (T1021.006)—allows investigators to better understand the behavior and intent of the attacker.

Task 5: Practical

This task involves a hand-on practice of placing terms in their correct definition. Once the user does this, they uncover a challenge flag. This is shown in the image below:



Upon the completion of the challenge, a capture flag is obtained. This is then used to answer the main question of the hands-on task. This is shown in the image below:



Key Takeaway

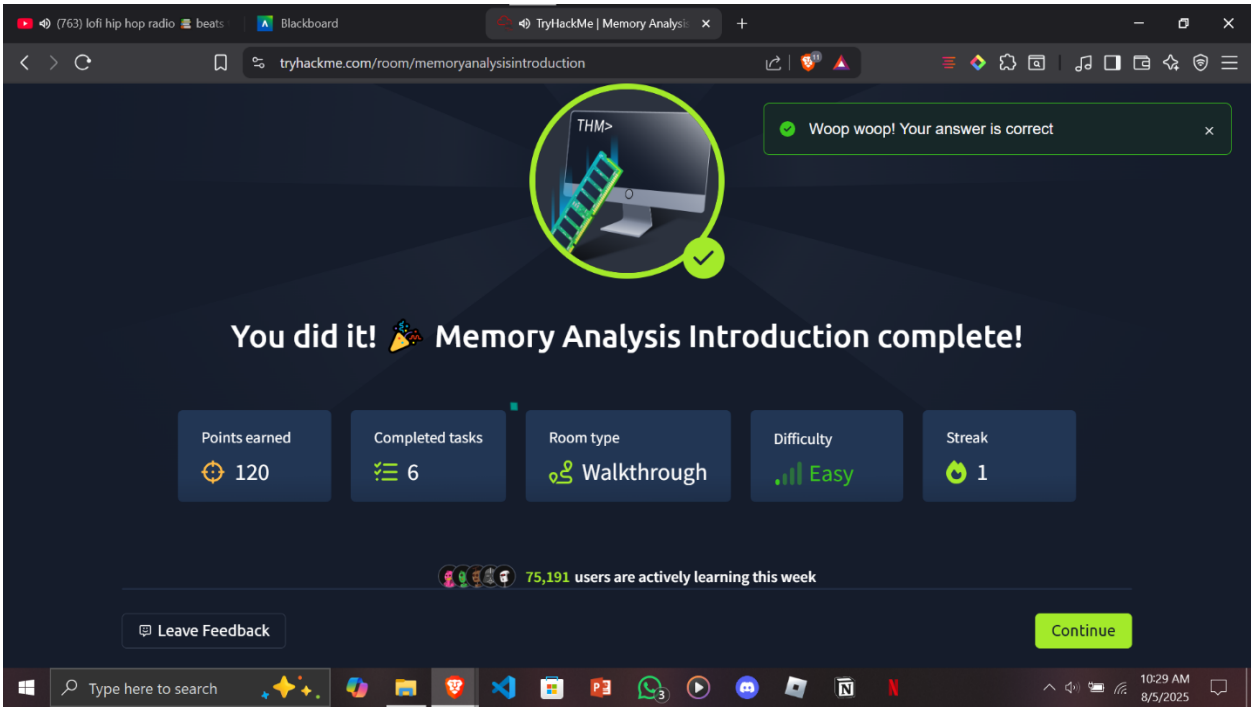
This final exercise provides hands-on experience in memory dump analysis to uncover the challenge flag **"THM{m3mory_analyst_g00d_job}".** It reinforces learned knowledge by guiding users through discovering command history, malicious processes, and artifacts like credentials stored in RAM. The practical exercise consolidates the theory learned, making users more comfortable with memory forensic tools and real-world investigative techniques.

Task 6: Conclusion

In this room, we leanred the importance of volatile memory analysis in computer forensics and how it contributes to the initiation phase of investigations. We explained what volatile memory is, why RAM stores running information, and why it's of top priority during incident response. We explained memory structure and how forensic data like commands, encryption keys, and processes get allocated in different memory regions.

We also went over memory dumps and how they are created, what forms they take, and the challenges in capturing them. Finally, we spoke about common attack techniques that can be identified using memory analysis, such as credential dumping, DLL injection, script execution, and signs of persistence or lateral movement.

All of these topics provide a strong foundation of the manner in which RAM is critical to threat detection.



References

TryHackMe. (n.d.). *Memory Analysis Introduction*.

https://tryhackme.com/room/memoryanalysisintroduction